

OPTICS & OPTICS LAB

7 – Course Schedule

No classes: Sept 7 (Labor Day Holiday), **Oct 12–13** (Fall Break), **Nov 11** (Veterans Day), **Nov 25–27** (Thanksgiving Holiday).

Topics & Concepts Chapters	Lab Experiments	Tests Final Project	HW Projects
HISTORICAL OVERVIEW Hecht 1 INTRODUCTORY OPTICS Wave Optics Knight 22 <i>Models of light; Double-slit interference; Diffraction grating; Single-slit diffraction; Circular-aperture diffraction; Interferometers</i>	Lab 1 <i>Laser & lab safety</i> <i>Optical components</i> <i>Laser alignment</i>		HW 1
Ray Optics Knight 23 <i>Ray model of light; Reflection; Refraction; Image formation; Dispersion & color; Thin lenses; Ray tracing</i>	Lab 2 <i>Beam expanders</i> <i>Laser beam profile</i>		
Optical Instruments Knight 24 <i>Lens systems; Cameras; Vision; Microscope; Telescope; Resolution</i>			HW 2
INTERMEDIATE OPTICS Wave Motion Hecht 2 <i>Harmonic waves (1D); Phase & phase velocity; Superposition principle; Complex representation; Phasors & addition of waves; Plane waves; Wave equation; Spherical & cylindrical waves; Twisted light</i>	Lab 3 <i>Measuring & modeling diffraction from slits</i>	Test 1	Project 1
Electromagnetic Theory Hecht 3 <i>Laws of electromagnetism; Maxwell's equations; Electromagnetic waves; Energy & momentum; Poynting vector; Irradiance; Photon theory of light; Squeezed light; Radiation pressure; Synchrotron radiation; Electric dipole radiation; Radiative atomic transitions; Light in bulk matter; Scattering & absorption; Dispersion &</i>	Lab 4 <i>Focal length of lens (multiple methods)</i>		HW 3

Caveat: This document is **not** a contract. Policies, procedures, projects, and schedules may change due to external circumstances or for pedagogical reasons.

Topics & Concepts Chapters	Lab Experiments	Tests Final Project	HW Projects
<i>optical constants; Electromagnetic-photon spectrum; Quantum fields & virtual photons</i>			
Propagation of Light Hecht 4 <i>Why the sky is blue; Scattering & interference; Reflection; Refraction; Fermat's principle; Mirages; Boundary conditions at interfaces; Fresnel equations; Total internal reflection; Evanescent waves; Optical properties of metals; Plasma oscillations; Color; QED</i>			HW 4
Geometrical Optics Hecht 5 & 6 <i>Refracting surfaces; Gaussian optics & thin lenses; Graphical ray-tracing; Finite imaging & magnification; Thin-lens combinations; Stops & pupils; $f/\#$, NA; Planar mirrors; Prisms; Fiber optics; Eyes, eyeglasses; Magnifiers; Microscopes; Cameras; Telescopes; Gravitational lensing; Matrix method analysis of optical systems</i>	Lab 5 Michelson interferometer Refractive index of air		Project 2
Superposition of Waves Hecht 7 <i>Algebraic & complex methods; Standing waves; Beats; Anharmonic & nonperiodic waves; Fourier analysis</i>		Test 2	
Polarization Hecht 8 <i>Linear, circular, elliptical; Photon angular momentum; Polarizers, Malus's law; Birefringence; Polarization processes; Waveplates</i>	Lab 6 Refractive index of prism (multiple methods)		Project 3
Modern Optics Hecht 13 <i>Lasers & laser light; LIGO; Laser-induced fusion; Spatial frequencies; Holography; Raman effect & spectroscopy; Nonlinear optics</i>		Final Project Presentation	